

Felipe Espinosa

Paris, France | +33 7 66 09 03 35 | felipeespinosa.13@outlook.com
felipe-espinosa.vercel.app | linkedin.com/in/felipe-espinosa13 | github.com/Pipe1213

SUMMARY

AI/ML Engineer seeking full-time roles in LLM Agents, RAG, and applied GenAI. Built an enterprise data analysis agent (LangGraph/LangChain) achieving 88% end-to-end QA and cutting SQL iteration time to 51s. Research background in generative AI for speech synthesis (TTS) with a first-author paper at SSW 2025 and open-source releases. Work authorization: France (post-study permit), available Feb 2026.

EDUCATION

Université Paris-Saclay

Master of Science in Computer Science - Artificial Intelligence Track

September 2023 – September 2025

Grade: 16.2/20

Universidad Industrial de Santander

Bachelor of Science (B.Sc.) in Electronic Engineering

April 2011 – December 2016

Grade: 3.9/5.0

EXPERIENCE

Capital Fund Management (CFM) | Paris, France

ML Engineer Intern — LLM Agents & RAG

March 2025 – September 2025

- Built a data analysis agent that achieved 88% end-to-end QA accuracy on an internal benchmark.
- Reduced average SQL writing and correction time to 51 seconds, speeding up analyst iterations and QA cycles.
- Achieved 95% top-k retrieval accuracy by combining BM25, dense embeddings, and RRF fusion in the RAG pipeline.
- Integrated multi-tool capabilities (coding, data visualization, and analytical reasoning) to enable complex, autonomous analyst workflows.
- Implemented active learning loops capturing user feedback for few-shot grounding and fine-tuning data curation.
- Deployed the system to a pilot group within the Data team to validate the architecture for future enterprise integration.

Laboratoire Interdisciplinaire des Sciences du Numérique (LISN) | Orsay, France

AI Scientist Intern — Speech Synthesis

February 2024 – August 2024

- Trained the first Text-to-Speech model for Walloon (3h of data) using cross-lingual transfer from 20h of French, achieving MOS 4.22/5 for the best configuration.
- Ran ablation studies (French pretraining vs none, graphemes vs phonemes) and conducted objective + human evaluation (PESQ, MCD, SECS + native-speaker perceptual tests) to select the best configuration.
- Lead author of a paper presented at the 13th Speech Synthesis Workshop (SSW 2025), in collaboration with senior researchers at LISN, highlighting cross-lingual transfer strategies for low-resource TTS.
- Open-sourced trained models, demos, and inference scripts on Hugging Face, enabling reuse by other researchers.

Independent Contractor | Sydney, Australia

Construction / Renovation Projects

March 2017 – July 2023

- Worked on residential and commercial renovation projects, delivering scoped tasks and supporting on-site execution.
- Coordinated with clients, supervisors, and trades to meet schedules, requirements, and safety standards.
- Ended contracting work in 2023 and relocated to France to pursue a specialized Master's degree in AI.

SKILLS

Programming: Python, SQL, Bash.

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, Matplotlib, NumPy, Transformers (HF), Jupyter.

Tools & Platforms: LangChain, LangGraph, Docker, Git, Slurm, Linux, Streamlit, OpenCV, ChromaDB, Redis.

Languages: Spanish (Native), English (C1), Portuguese (B2), French (B1).

PROJECTS

Structure from Motion 3D Reconstruction

February 2025

Implemented a full SfM pipeline with SIFT, FLANN, and PnP to reconstruct 3D scenes from RGB images; achieved accurate alignment on multi-view datasets using custom feature-matching logic with OpenCV.

Creation of a Synthetic Dataset for French NER

January 2024 – February 2024

Built a cross-lingual NER pipeline with Hugging Face Transformers; translated English CHIA data and fine-tuned XLM-RoBERTa for French annotation, evaluating translation quality with BLEU, TER, and METEOR metrics.